IMEN266 Operations Research II
Chapter 4. Markov Chains

Fall, 2026



INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

Outline

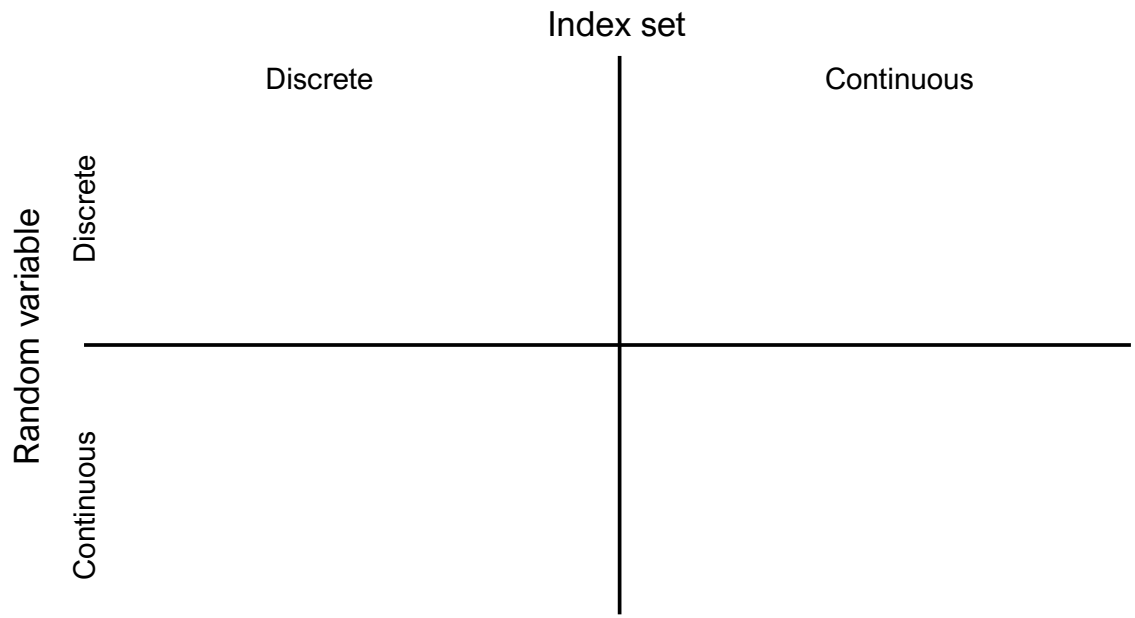
1. Stochastic process
2. Modeling with a Markov chain
3. Transient behavior
4. Limiting behavior
5. Cost and reward
6. PageRank practicum

How to study this chapter (2026 format)

- **Slides** (this file): concepts, class-problem statements, and the worked derivations developed in class. No transcription needed — definitions, theorems and proofs also live in the Companion.
- **Notebooks** (CP3 modeling · CP4 transient · CP5 limiting/cost · PageRank + handout PDF): in class the `INCLASS` version (fill the `TODO`s, `check()` answers instantly); at home the full version with solutions, sliders, experiments.
- **Companion** (gap-fill → full): all definitions and theorems + 8 core proofs with blanks, checkpoints, AI study cards.
- **Homework 2**: A exam-style (no AI first) · B critique an AI solution · C stress-test the Markov assumption · D prompt log.
- Class rhythm (75 min): quiz $5'$ → concepts → live notebook → wrap-up. Ch. 4 = weeks 3–5.
- Ground truth = analytic $\leftrightarrow$ simulation agreement — not AI confidence. Materials: PLMS and GitHub `imen266-2026`.

What is it?

- Stochastic + Process: $\{X_t, t \in \mathcal{I}\}$.
- X_t: random variable for a fixed t.
- $\mathcal{I}$: index set



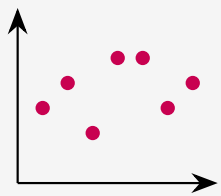
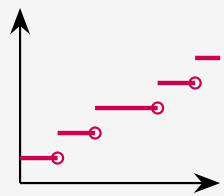
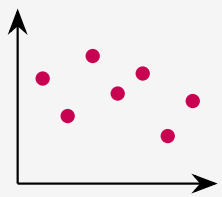
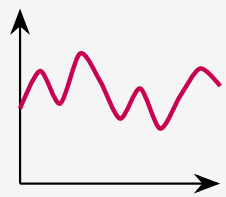
▷ Companion §1.1: the taxonomy table — filled in during the CP3 warm-up.

What is it?

Reading the definition

Stochastic ← uncertainty (probability); process ← time (index). A stochastic process is a family of *indexed random variables* $\{X_t,\ t \in \mathcal{I}\}$.

The four quadrants, with examples

random variable \ **index set**:	discrete	continuous
discrete	**Ch. 4 Markov chains** inventory at the start of each day 	**Ch. 5–7**: Poisson process, CTMC, renewal e-mail arrivals 
continuous	**Time-series analysis** temperature at the start of each hour 	**Brownian motion** (Intro. to Financial Engineering) 

Class problem 3

1. A company manufactures a machine that is either up or down. If it is up at the beginning of a day, then it is up at the beginning of the next day with probability 0.98 (regardless of the history of the machine); or it fails with probability 0.02. Once the machine goes down, the company sends a repair person to repair it. If the machine is down at the beginning of a day, it is down at the beginning of the next day with probability 0.03 (regardless of the history of the machine); or the repair completes and the machine is up with probability 0.97. A repaired machine is as good as new. Let X_n be the state of the machine at the beginning of the nth day. Model $\{X_n, n \geq 0\}$ as a DTMC.

▶ Colab notebook: CP3 §1 (machine up/down) `[INCLASS] [full]`

Class problem 3

1. Machine up or down: the three things every DTMC model needs

- $X_n = 0$ if the machine is down at the beginning of day n, $X_n = 1$ if it is up.
  $X_n \equiv$ state of the system at index n
- $S = \{0, 1\}$ $S \equiv$ state space: the set of all values X_n can take
- Transition probabilities read off the story ("regardless of the history" is the Markov property):

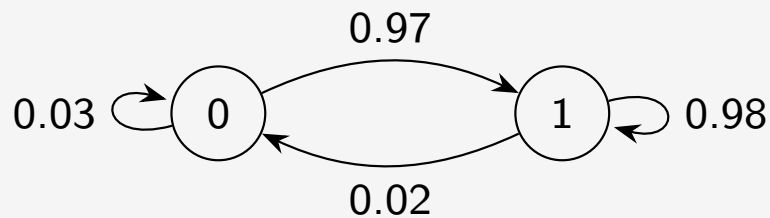
$$P(X_{n+1}{=}1 \mid X_n{=}1) = 0.98, \quad P(X_{n+1}{=}0 \mid X_n{=}1) = 0.02,$$
$$P(X_{n+1}{=}1 \mid X_n{=}0) = 0.97, \quad P(X_{n+1}{=}0 \mid X_n{=}0) = 0.03.$$

$P \equiv$ transition probability matrix

$$P = \begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 0.03 & 0.97 \\ 1 & 0.02 & 0.98 \end{array}$$

row: from
column: to

Transition diagram



Markov chains (MCs, DTMCs)

- $\{X_n, n \in \mathbb{Z}+\}$ is called a Markov chain if it satisfies the following condition:

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0)$$

$$= P(X_{n+1} = j | X_n = i)$$

$$= P(X_1 = j | X_0 = i).$$

▷ Companion §1.2 and Proof 1: Markov property ⇒ path probabilities multiply.

Markov chains (MCs, DTMCs)

Two separate assumptions hide in the definition

$$P(X_{n+1} = j \mid X_n = i, \underbrace{X_{n-1} = i_{n-1}, \ \ldots, \ X_0 = i_0}_{\text{ignored}})$$

$$\overset{(1)}{=} P(X_{n+1} = j \mid X_n = i) \overset{(2)}{=} P(X_1 = j \mid X_0 = i) =: p_{ij}$$

(1) **Markov property**: ignore the entire history other than X_n.

(2) **Time homogeneity** (stationary transition probabilities): the one-step probability does not depend on n. The p_{ij} are collected in the matrix P.

Without time homogeneity

the one-step probabilities would carry a time index:
$p_{ij}(n) = P(X_{n+1} = j \mid X_n = i)$.

Modeling a system as a MC

1. Define X_n, the state at n^{th} observation.
2. Write down $\mathcal{S}$, the state space.
3. Verify Markov & time-homogeneity properties
4. Derive P, the transition probability matrix.
5. Draw a transition diagram.

▷ Companion §1.2 (recipe). Every CP3 problem follows these five steps; `DTMC(P, states)` checks the row sums.

Class problem 3

2. The weather in the city of Pohang is classified as sunny, cloudy or rainy. Suppose that tomorrow's weather depends only on today's weather as follows: If it is sunny today, it is cloudy tomorrow with probability 0.3 and rainy with probability 0.2; if it is cloudy today, it is sunny tomorrow with probability 0.5 and rainy with probability 0.3; and finally, if it is rainy today, it is sunny tomorrow with probability 0.4 and cloudy with probability 0.5. Let X_n be the type of the nth day. Model $\{X_n, n \geq 0\}$ as a DTMC.

▶ Colab notebook: CP3 §2 (Pohang weather) `[INCLASS] [full]`

Class problem 3

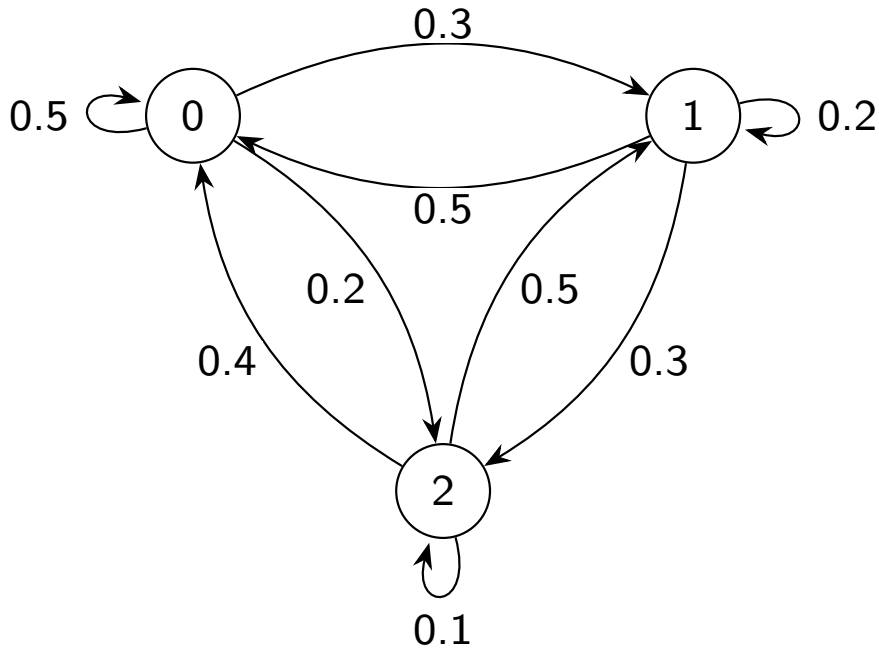
2. Weather in Pohang

$$X_n = \begin{cases} 0 & \text{sunny on day } n \\ 1 & \text{cloudy} \\ 2 & \text{rainy} \end{cases},$$

$S = \{0, 1, 2\}$

$$P = \begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0.5 & 0.3 & 0.2 \\ 1 & 0.5 & 0.2 & 0.3 \\ 2 & 0.4 & 0.5 & 0.1 \end{array}$$

Each row must add to 1: that is how the "stays the same" probabilities are filled in.



Class problem 3

3. Himart stocks a wide variety of PCs for retail sales. It is open for business Monday through Friday 8am to 5pm. It uses the following operating policy to control the inventory of Paint-yum PCs: At 5pm Friday, the store clerk checks to see how many Paint-yum PCs are still in stock. If the number is less than 2, then he orders enough Paint-yum PCs to bring the stock up to 5 at the beginning of the business day Monday. If the number in stock is 2 or more, no action is taken. The demand for Paint-yum PCs during the week is a Poisson random variable with mean 3. Any demand that cannot be immediately satisfied is lost. Develop a DTMC to model the inventory of Paint-yum PCs at Himart. For your convenience, use the table below for weekly demand distribution:

k	0	1	2	3	4
$P[\text{Demand} = k]$	0.0498	0.1494	0.2240	0.2240	0.1680
$P[\text{Demand} \geq k]$	1.0000	0.9502	0.8008	0.5768	0.3528

▶ Colab notebook: CP3 §3 (Himart inventory) `[INCLASS] [full]`

Class problem 3

3. Himart: list the transition *events* first, then the probabilities

$X_n \equiv$ number of PCs in inventory at the start of the n-th Monday;
$S = \{2, 3, 4, 5\}$.

demand D moving $i \to j$	$j=2$	3	4	5
$i=2$	D=0	–	–	$D \geq 1$
3	D=1	D=0	–	$D \geq 2$
4	D=2	D=1	D=0	$D \geq 3$
5	D=3	D=2	D=1	$D \geq 4$ or D=0

Then read Poisson(3) off the table

$$P = \begin{pmatrix} 0.0498 & 0 & 0 & 0.9502 \\ 0.1494 & 0.0498 & 0 & 0.8009 \\ 0.2240 & 0.1494 & 0.0498 & 0.5768 \\ 0.2240 & 0.2240 & 0.1494 & 0.4026 \end{pmatrix}$$

row $i = 2$ is worked out in class; the gray rows follow the same recipe — check them, and draw the transition diagram, yourself

Class problem 3

4. There are two identical routes for messages to traverse from node A to node B. If a route was congested when a message needed to be sent, then it will be congested with probability q when the next message needs to be sent. Likewise, if a route was congestion-free when a message needed to be sent, then it will be congestion-free with probability p when the next message needs to be sent. The two routes behave independently. Let X_n be the number of routes available when the nth message is attempted. Show that $\{X_n, n \geq 0\}$ is a DTMC.

▶ Colab notebook: CP3 §4 (two routes) `[INCLASS] [full]`

Class problem 3

4. Two routes from A to B (each congested or congestion-free, independently)

$X_n \equiv$ number of routes available when the n-th message is attempted, $S = \{0, 1, 2\}$.

$$P = \begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & q^2 & 2q(1-q) & (1-q)^2 \\ 1 & q(1-p) & (1-q)(1-p)+qp & p(1-q) \\ 2 & (1-p)^2 & 2p(1-p) & p^2 \end{array}$$

Row 1: one route is congested (stays so w.p. q), the other is free (stays so w.p. p); combine the two independent moves. Transition diagram: exercise.

Class problem 3

5. An individual possesses r umbrellas, which she employs in going from her home to office and vice versa. If she is at home (the office) at the beginning (end) of a day and its raining, then she will take an umbrella with her to the office (home), provided there is one to be taken. If it is not raining, then she never takes an umbrella. Assume that, independent of the past, is rains at the beginning (end) of a day with probability p. Define a DTMC with $r+1$ states that will help us determine the proportion of time that our individual gets wet. Note: She gets wet if it is raining and all the umbrellas are at her other location.

▶ Colab notebook: CP3 §5 (umbrellas) `[INCLASS] [full]`

Class problem 3

5. Umbrellas: $X_n \equiv$ number of umbrellas at the location where she begins her n-th trip

$S = \{0, 1, \ldots, r\}$. Starting with y umbrellas here means $r - y$ at the other place; after the trip she starts from there with $r - y$ (no rain) or $r - y + 1$ (rain: she carried one over). From 0 she must arrive to r.

$$P = \begin{array}{c|cccccc} & 0 & 1 & 2 & \cdots & r-1 & r \\ \hline 0 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & 0 & \cdots & 1-p & p \\ 2 & 0 & 0 & 0 & 1-p & p & 0 \\ \vdots & & & \ddots & & & \\ r-1 & 0 & 1-p & p & \cdots & 0 & 0 \\ r & 1-p & p & 0 & \cdots & 0 & 0 \end{array}$$

In symbols: $p_{i,\,r-i} = 1 - p$ and $p_{i,\,r-i+1} = p$ for $i \geq 1$; $p_{0,r} = 1$.
Transition diagram: exercise.

Class problem 3

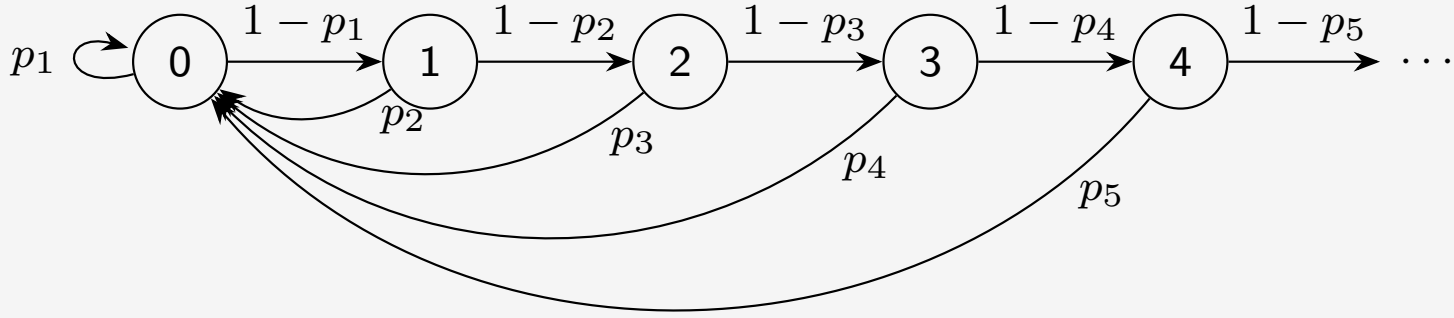
6. Consider a machine tool that is used on an NC-machine. The time to produce a part and inspect it is 1-hour. If the tool has produced i parts, it fails with probability p_i. Let X_n be the age of a machine tool at the beginning of nth hour. Assuming that a failed tool is replaced by a new tool whenever it fails, model $\{X_n, n \geq 0\}$ as a DTMC.

▶ Colab notebook: CP3 §6 (tool age) `[INCLASS] [full]`

Class problem 3

6. Machine tool: $X_n \equiv$ age of the tool at the beginning of the n-th hour, $S = \{0, 1, 2, \ldots\}$

During each hour the tool produces one part and it is inspected; a failed tool is replaced by a new one (age 0).



$p_{i,i+1} = 1 - p_{i+1}$ (survives the next part), $p_{i,0} = p_{i+1}$ (fails and is replaced). P: exercise.

Class problem 3

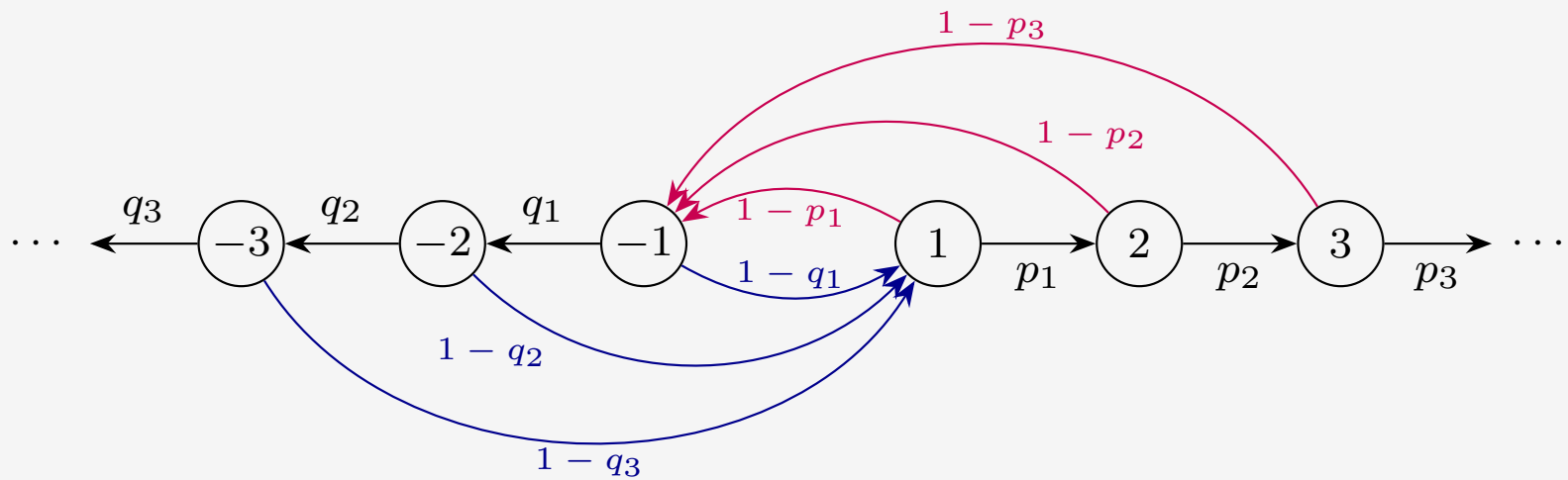
7. Consider the following weather forecasting model: if today is sunny (rainy) and it is the kth day of the current sunny (rainy) spell, then it will be sunny (rainy) tomorrow with probability $p_k(q_k)$ regardless of what happened before the current sunny (rainy) spell started. Model this as a DTMC.

- Colab notebook: CP3 §7 (weather spells) `[INCLASS] [full]`

Class problem 3

7. Sunny and rainy spells: a first attempt, and the state that works

A first idea, $X_n = (X_n^1, X_n^2)$, carries more than we need. Instead keep the sign for the type and the magnitude for the day within the spell: $X_n = i$ if day n is the i-th day of a sunny spell, $X_n = -i$ if it is the i-th day of a rainy spell; $S = \{\ldots, -2, -1, 1, 2, \ldots\}$.



Sunny day i: continue to $i + 1$ w.p. p_i, else a rainy spell starts (-1).
Rainy day i: continue to $-(i + 1)$ w.p. q_i, else a sunny spell starts (1).

Class problem 4

1. Consider the 3-state DTMC for weather in the city of Pohang in the previous class problem. Answer the following questions (all independent):
   (a) According to weather forecast today will be rainy with probability 0.3 and sunny with probability 0.4. What is the probability it would be cloudy tomorrow? How about day after? What about 3 days from today?
   (b) You know that today is sunny. What is the probability that it would be rainy day after tomorrow?
   (c) If yesterday was sunny, day before was sunny and today is rainy, what is the probability that it would be cloudy day after tomorrow?

▶ Colab notebook: CP4 §1 (weather, transient) `[INCLASS] [full]`

Class problem 4

1(a): tomorrow, by the law of total probability

Today's forecast (0 sunny, 1 cloudy, 2 rainy):
$P(X_0{=}0) = 0.4,\ P(X_0{=}1) = 0.3,\ P(X_0{=}2) = 0.3.$

$$\begin{aligned} P(X_1{=}1) &= P(X_1{=}1 \mid X_0{=}0)\,P(X_0{=}0) + P(X_1{=}1 \mid X_0{=}1)\,P(X_0{=}1) \\ &\quad + P(X_1{=}1 \mid X_0{=}2)\,P(X_0{=}2) \\ &= 0.3(0.4) + 0.2(0.3) + 0.5(0.3) = 0.33. \end{aligned}$$

All three states at once: a row vector times P

$$a \equiv \begin{bmatrix} P(X_0{=}0) & P(X_0{=}1) & P(X_0{=}2) \end{bmatrix} \quad \text{(initial distribution)}$$

$$p_1 \equiv \begin{bmatrix} P(X_1{=}0) & P(X_1{=}1) & P(X_1{=}2) \end{bmatrix} = aP,$$

$$p_2 = p_1 P = aP^2, \qquad p_3 = p_2 P = aP^3, \qquad \ldots, \qquad p_n = aP^n.$$

The entries of P^n are the n-step probabilities $P(X_n{=}j \mid X_0{=}i)$. Day after tomorrow: $(aP^2)_1 = 0.307$; three days from today: $(aP^3)_1 = 0.312$.

Class problem 4

1(b), 1(c): the same machinery with a different a

(b) Today is sunny: $a = (1\ 0\ 0)$, $P(X_2 = 2) = (aP^2)_2 = 0.21$.

(c) Yesterday and the day before were sunny, today is rainy: only *today* matters (Markov property), so $a = (0\ 0\ 1)$ and $P(X_2 = 1) = (aP^2)_1 = 0.27$.

Transient analysis

- Initial distribution

$$a \equiv [P(X_0 = 0), P(X_0 = 1), \ldots].$$

- Define

$$p_n \equiv [P(X_n = 0), P(X_n = 1), \ldots].$$

- Transient analysis

$$\begin{aligned} p_1 &= aP \\ p_2 &= p_1 P = aP^2 \\ &\vdots \\ p_n &= aP^n. \end{aligned}$$

▷ Derivation: Companion, Proof 2 (Chapman–Kolmogorov, $p_n = aP^n$).

Class problem 4

3. Consider a DTMC with initial distribution a given by $a = (0.3, 0.5, 0.1, 0.1)$ and transition probability matrix P given by

$$P = \begin{pmatrix} 0.25 & 0.25 & 0.25 & 0.25 \\ 0.1 & 0.2 & 0.3 & 0.4 \\ 0.6 & 0.1 & 0.1 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.4 \end{pmatrix}.$$

Compute the following:

(a) $P[X_3 = 3, X_2 = 3, X_1 = 2, X_0 = 1]$
(b) $P[X_3 = 3, X_2 = 2, X_1 = 1]$
(c) $P[X_5 = 2 | X_1 = 3]$
(d) $P[X_3 = 1]$
(e) Distribution of X_5
(f) $E[X_4]$
(g) Var$[X_4]$

▶ Colab notebook: CP4 §3 (4-state DTMC) `[INCLASS] [full]`

Class problem 4

3(a): a path probability — peel off one step at a time (Markov property)

$$
\begin{aligned}
&P(X_3{=}3, X_2{=}3, X_1{=}2, X_0{=}1) \\
&\quad = P(X_3{=}3 \mid X_2{=}3, \cancel{X_1{=}2, X_0{=}1})\, P(X_2{=}3, X_1{=}2, X_0{=}1) \\
&\quad = P(X_3{=}3 \mid X_2{=}3)\, P(X_2{=}3 \mid X_1{=}2, \cancel{X_0{=}1})\, P(X_1{=}2, X_0{=}1) \\
&\quad = P(X_3{=}3 \mid X_2{=}3)\, P(X_2{=}3 \mid X_1{=}2)\, P(X_1{=}2 \mid X_0{=}1)\, P(X_0{=}1) \\
&\quad = p_{33}\, p_{23}\, p_{12}\, a_1 \quad = (0.1)(0.3)(0.25)(0.3) = 0.00225.
\end{aligned}
$$

Each step: condition on the most recent state only, then multiply by the probability of the shorter path.

Class problem 4

3(b)–(g), one line each

(b) exercise — same pattern as (a) $(= (aP)_1\, p_{12}\, p_{23} = 0.0154)$

(c) $P(X_5{=}2 \mid X_1{=}3) = \left[P^4\right]_{32} = 0.193$

(d) $P(X_3{=}1) = \left[aP^3\right]_1 = 0.278$

(e) distribution of X_5: $p_5 = \left[P(X_5{=}1) \;\cdots\; P(X_5{=}4)\right] = aP^5$

(f) $E[X_4] = \sum_{i=1}^{4} i\, P(X_4{=}i)$, with $P(X_4{=}i)$ read from aP^4 $(= 2.564)$

(g) $\mathrm{Var}(X_4) = E[X_4^2] - E[X_4]^2$ $(= 1.436)$

Class problem 4

2. Consider the 4-state DTMC for the Himart inventory PCs problem in the previous class problems. Assume that the business starts with 5 Paint-yum PCs in hand at the beginning. Answer the following questions (all independent):
   (a) What is the initial distribution a?
   (b) What is the probability that at the end of that week Paint-yum PCs have to be ordered for the next week?
   (c) What is the probability that after 4 weeks there are 3 Paint-yum PCs?
   (d) If you are given that at the beginning of the 17th week there are 3 Paint-yum PCs, what is the probability that at the beginning of the 20th week there would be 5?

▶ Colab notebook: CP4 §2 (Himart, transient) `[INCLASS] [full]`

Class problem 4

2. Himart, $S = \{2, 3, 4, 5\}$, start with 5 PCs

(a) $a = (0 \;\; 0 \;\; 0 \;\; 1)$.

(b) An order is placed when fewer than 2 remain: $P(\text{demand} \geq 4)$ from the table $= 0.3528$.

(c) $P(X_4 = 3) = \left[aP^4\right]_3$ — the column of state 3 in aP^4 $(= 0.147)$.

(d) $P(X_{20} = 5 \mid X_{17} = 3) = P(X_3 = 5 \mid X_0 = 3)$: three steps, so read row 3, column 5 of P^3 $(= 0.602)$.

Important properties

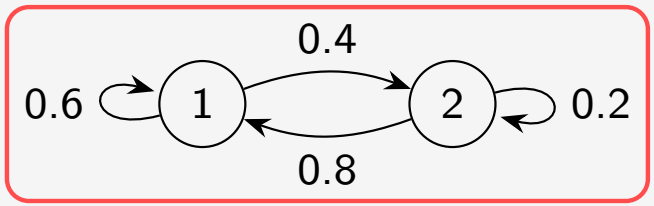
$$P = \begin{pmatrix} 0.6 & 0.4 & 0 & 0 & 0 \\ 0.8 & 0.2 & 0 & 0 & 0 \\ 0 & 0 & 0.3 & 0.2 & 0.5 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0.8 & 0.1 & 0.1 \end{pmatrix}$$

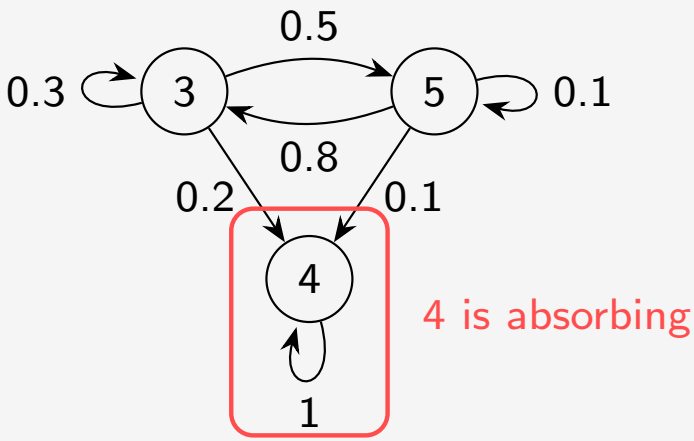
$$P = \begin{pmatrix} 0.6 & 0.4 & 0 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0.5 & 0.5 \end{pmatrix}$$

Important properties

Draw the diagram first — the classes jump out

first P:

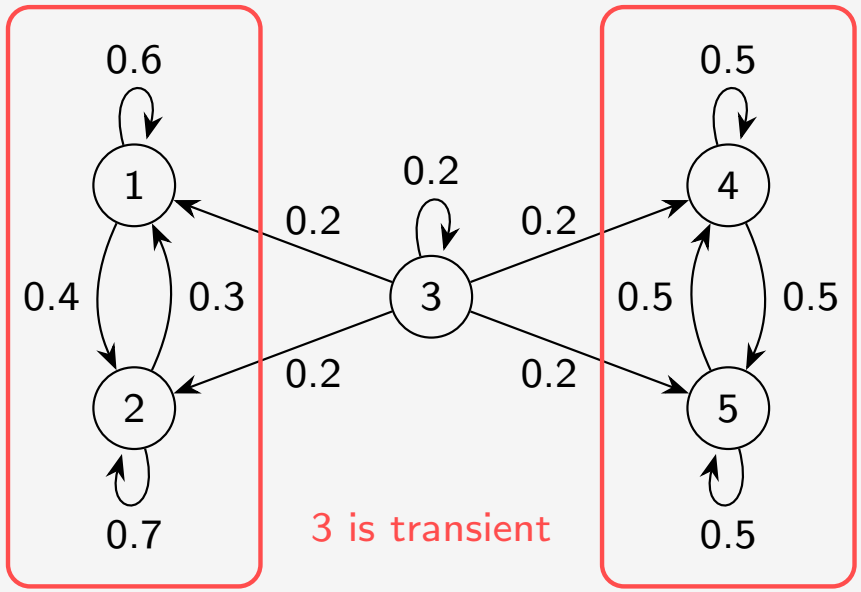




4 is absorbing

closed: $\{1,2\}$, $\{4\}$; $3,5$ leak into 4.

second P:



3 is transient

closed: $\{1,2\}$, $\{4,5\}$; 3 leaves and never returns.

So $\lim_{n\to\infty} P(X_n = j)$ depends on where you start — the reducible case.

Important properties

- Irreducible vs. reducible
- Recurrent vs. transient
- Aperiodic vs. periodic

▷ Definitions: Companion §1.4; Proof 3 (recurrence criterion, class property); `DTMC.summary()` classifies for you.

Important properties

1. Irreducible: "you can go everywhere"

For all $i, j \in S$ there exists $n > 0$ such that $P(X_n = j \mid X_0 = i) > 0$.

2. Recurrent: "you can come back for sure"

Let $T_j \equiv \inf\{n > 0 : X_n = j\}$ (first return time). Recurrent: $P(T_j < \infty \mid X_0 = j) = 1$.
Positive recurrent: $E[T_j \mid X_0 = j] < \infty$; *null* recurrent: $E[T_j \mid X_0 = j] = \infty$.
A certain return can still have infinite mean — example: $P(T = n) = \frac{1}{n(n+1)}$:

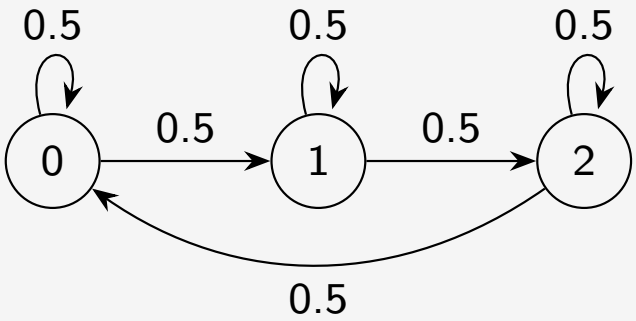
$$\sum_{n\geq 1} \frac{1}{n(n+1)} = 1 \quad \text{but} \quad E[T] = \sum_{n\geq 1} \frac{n}{n(n+1)} = \sum_{n\geq 1} \frac{1}{n+1} = \infty.$$

3. Aperiodic: "no cycle in the occurrence of a state"

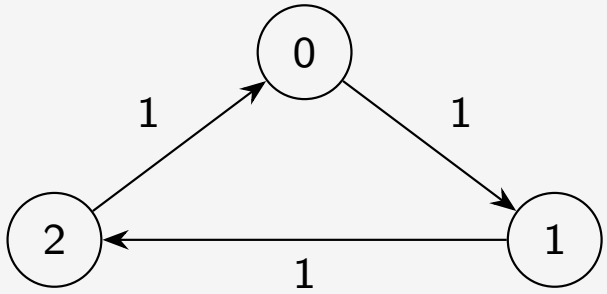
Period $d =$ smallest number such that state j can only recur at integer multiples of d. If $d = 1$ the chain is aperiodic.

Important properties

Periodic or not? Two three-state examples



self-loops $\Rightarrow d = 1$: aperiodic



returns only at times $3, 6, 9, \ldots \Rightarrow$ $d = 3$: periodic

Desired Markov chains (Ergodic)

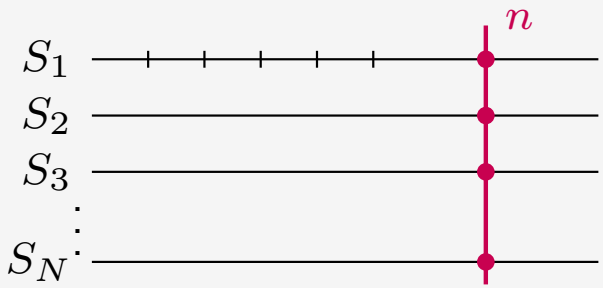
- Irreducible
- Positive recurrent
- Aperiodic

▷ Companion §1.4: ergodic = irreducible + positive recurrent + aperiodic. Proof 7: why taxation makes PageRank ergodic.

Desired Markov chains (Ergodic)

Ergodic = the limit exists, and *ensemble average* = *time average*

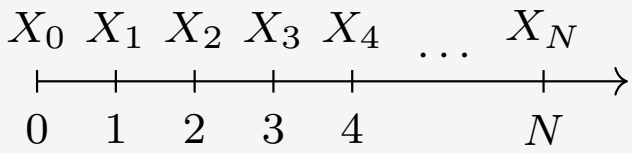
1) Ensemble: N independent copies, all read at time n



ensemble average

$$\frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\{X_n^i = j\}$$

2) Time: one path, the fraction of time spent in j



time average $\dfrac{1}{N+1}\sum_{i=0}^{N}\mathbf{1}\{X_i = j\}$

For an ergodic chain both averages $\approx \lim_{n\to\infty} P(X_n = j) = \pi_j$ for large N. "Sometimes you interpret this probability as the fraction of time that the system is in state j" — that interpretation is exactly what ergodicity buys.

Limiting behavior

- If a Markov chain is irreducible and positive recurrent, we have a *unique solution* to

$$\boldsymbol{\pi} = \boldsymbol{\pi} P, \quad \sum_{i \in \mathcal{S}} \pi_i = 1,$$

  where $\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots)$.
- If the Markov chain above is aperiodic,

$$\lim_{n \to \infty} P(X_n = i) = \pi_i.$$

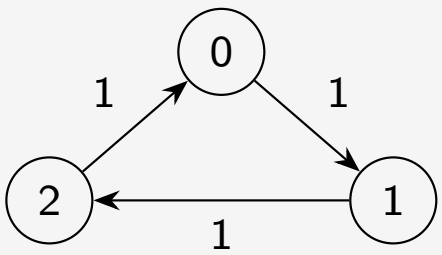
- Example

▷ Contrast with transient analysis, $p_1 = aP$ — there the answer depends on a; here it does not.

▷ Companion §1.5; Proof 4 (two-state $P^n \to \pi$), Proof 5 (solving $\pi = \pi P$), Proof 6 ($\pi_i = 1/m_i$).

Limiting behavior

Example: a stationary distribution can exist even when the limit does not



$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Irreducible, positive recurrent, but *periodic* ($d = 3$).

$\pi = \pi P$: $\pi_0 = \pi_2,\ \pi_1 = \pi_0,\ \pi_2 = \pi_1$; with $\sum_i \pi_i = 1$, $\pi_0 = \pi_1 = \pi_2 = \frac{1}{3}$.

The unique solution of $\pi = \pi P$ exists, but from a fixed start $P(X_n = i)$ keeps cycling through the values 0 and 1 (from $X_0 = 0$: $P(X_n = 0) = 1, 0, 0, 1, 0, 0, \ldots$), so $\lim_n P(X_n = i)$ does not exist. Aperiodicity is what turns π into a limit.

Class problem 5

1. Consider problem #1 in Class Problems 3 on DTMC modeling. If it is up at the beginning of a day, then it is up at the beginning of the next day with probability 0.98 (regardless of the history of the machine); or it fails with probability 0.02. Once the machine goes down, the company sends a repair person to repair it. If the machine is down at the beginning of a day, it is down at the beginning of the next day with probability 0.03 (regardless of the history of the machine); or the repair completes and the machine is up with probability 0.97. For this system, compute the steady state probabilities that the machine is up and down at the beginning of a day in the long run.

▶ Colab notebook: CP5 §1 (machine, steady state) `[INCLASS] [full]`

Class problem 5

1. Machine up/down in the long run

Finite state space + irreducible $\Rightarrow$ positive recurrent, so $\pi = \pi P$ has a unique solution.

$$(\pi_0 \quad \pi_1) = (\pi_0 \quad \pi_1) \begin{pmatrix} 0.03 & 0.97 \\ 0.02 & 0.98 \end{pmatrix}, \quad \pi_0 + \pi_1 = 1$$

$$\pi_0 = 0.0202 = \lim_{n\to\infty} P(X_n{=}0), \quad \pi_1 = 0.9798 = \lim_{n\to\infty} P(X_n{=}1).$$

What P^n looks like for large n

$$P^n \xrightarrow[n\to\infty]{} \begin{pmatrix} \pi_0 & \pi_1 \\ \pi_0 & \pi_1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{\pi} \\ \boldsymbol{\pi} \end{pmatrix} \qquad \text{every row is } \boldsymbol{\pi}.$$

The starting state is forgotten: $P(X_n = j \mid X_0 = i) \to \pi_j$ for every i.

Class problem 5

2. Consider problem #2 in Class Problems 3 on DTMC modeling about the weather being one of sunny, cloudy or rainy everyday. If it is sunny today, it is cloudy tomorrow with probability 0.3 and rainy with probability 0.2; if it is cloudy today, it is sunny tomorrow with probability 0.5 and rainy with probability 0.3; and finally, if it is rainy today, it is sunny tomorrow with probability 0.4 and cloudy with probability 0.5. What fraction of days would be sunny, cloudy or rainy in this city?

▶ Colab notebook: CP5 §2 (weather, steady state) `[INCLASS] [full]`

Class problem 5

2. Weather in the long run

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.5 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.1 \end{pmatrix}, \qquad \pi = \pi P, \quad \sum_i \pi_i = 1$$

$$\Rightarrow \quad \pi_0 = \frac{2.28}{4.76}, \qquad \pi_1 = \frac{1.48}{4.76}, \qquad \pi_2 = \frac{1}{4.76}.$$

$\boldsymbol{\pi} \approx (0.479,\ 0.311,\ 0.210)$: about 48% of the days sunny, 31% cloudy, 21% rainy.

Class problem 5

3. Consider problem #5 in Class Problems 3 on DTMC modeling about this lady with r umbrellas. Compute the limiting probabilities $(\pi_0, \pi_1, \ldots, \pi_r)$ and also determine the fraction of trips the individual becomes wet (φ). Every time the individual becomes wet, it costs c won to recover (medication, new clothes, or whatever). Also the maintenance cost for r umbrellas is d won per umbrella per trip (basically to purchase new umbrellas to replace old ones). Let τ be the long-run average cost per trip, then $\tau = c\varphi + dr$. In terms of p, c and d, determine the optimal number of umbrellas the individual should possess.

▶ Colab notebook: CP5 §3 (umbrellas, cost) `[INCLASS] [full]`

Class problem 5

3. Umbrellas: solve $\pi = \pi P$ column by column

$$\pi_0 = (1-p)\,\pi_r, \qquad \pi_1 = (1-p)\,\pi_{r-1} + p\,\pi_r,$$
$$\pi_2 = (1-p)\,\pi_{r-2} + p\,\pi_{r-1}, \quad \dots \qquad \pi_r = \pi_0 + p\,\pi_1.$$

Substitute π_0 into the last equation: $\pi_r = (1-p)\pi_r + p\pi_1 \Rightarrow \pi_1 = \pi_r$. Then $\pi_1 = (1-p)\pi_{r-1} + p\pi_r$ with $\pi_1 = \pi_r$ gives $\pi_{r-1} = \pi_r$, and so on: $\pi_1 = \pi_2 = \cdots = \pi_r$. Normalize: $\pi_0 + r\,\pi_r = 1$, so

$$\boldsymbol{\pi} = \Big(\frac{1-p}{1-p+r},\ \frac{1}{1-p+r},\ \dots,\ \frac{1}{1-p+r}\Big).$$

Class problem 5

3, continued: how many umbrellas should she own?

Fraction of trips she gets wet: $\varphi = \pi_0 \cdot p$ (no umbrella where she is, and it rains).

$$\tau = c\,\varphi + d\,r = c\,\pi_0\,p + d\,r = c \cdot \frac{1-p}{1-p+r} \cdot p + d\,r$$

$$\frac{d\tau}{dr} = -\frac{c\,p(1-p)}{(1-p+r)^2} + d = 0 \quad \Rightarrow \quad r^* = -1 + p + \sqrt{\frac{c\,p\,(1-p)}{d}}.$$

(Then round to a neighbouring integer and compare τ.)

Cost and reward

- If a cost $C(i)$ is incurred every time the Markov chain is observed in state i, then the long-run average cost per observation is

$$\sum_{i \in \mathcal{S}} C(i)\pi_i.$$

▷ Proof 6 in the Companion: time average $\to \sum_i C(i)\pi_i$; mean return time $1/\pi_i$.

Class problem 5

4. Consider problem #6 in Class Problems 3 on DTMC modeling where if a machine tool produced i parts, it fails with probability p_i. Let X_n be the age of a machine tool at the beginning of nth hour. Compute the steady state distribution for the DTMC. What is the average age of a tool when it starts processing a part? What is the average lifetime of a tool?

- Colab notebook: CP5 §4 (tool age, steady state) `[INCLASS] [full]`

Class problem 5

4. Machine tool: $\pi = \pi P$ solves recursively

$$P = \begin{array}{c|cccc} & 0 & 1 & 2 & 3 & \cdots \\ \hline 0 & p_1 & 1-p_1 & 0 & 0 & \cdots \\ 1 & p_2 & 0 & 1-p_2 & 0 & \cdots \\ 2 & p_3 & 0 & 0 & 1-p_3 & \cdots \\ 3 & p_4 & 0 & 0 & 0 & \ddots \end{array}$$

Column 0 of $\pi = \pi P$ is the only long equation; every other column has a single term:

$$\pi_0 = p_1\pi_0 + p_2\pi_1 + p_3\pi_2 + \cdots$$

$$\pi_1 = (1-p_1)\pi_0, \qquad \pi_2 = (1-p_2)\pi_1 = (1-p_2)(1-p_1)\pi_0,$$

$$\pi_3 = (1-p_3)\pi_2 = (1-p_3)(1-p_2)(1-p_1)\pi_0, \quad \ldots$$

$$\sum_{i\geq 0}\pi_i = 1 \;\Rightarrow\; \pi_0 = \frac{1}{1 + (1-p_1) + (1-p_1)(1-p_2) + \cdots}$$

Class problem 5

4, continued: two averages that are easy to confuse

Average age of the tool in use: reward $C(i) = i$ per observation, so

$$\text{average age} = \sum_{i=0}^{\infty} i\,\pi_i.$$

Average lifetime of a tool: a geometric-type distribution. With $X \equiv$ lifetime (in hours): $P(X{=}1) = p_1$, $P(X{=}2) = (1-p_1)\,p_2$, and in general $P(X{=}k) = (1-p_1)\cdots(1-p_{k-1})\,p_k$ (survives $k-1$ inspections, fails at the k-th), so

$$\text{average lifetime} = E[X] = \sum_{k=1}^{\infty} k\,P(X{=}k) \qquad (\text{not } \textstyle\sum_i i\,\pi_i).$$

▷ Tail sum: $E[X] = \sum_{i\geq 0} P(X > i) = 1/\pi_0$ (mean return time to 0).

Class problem 5

5. Kim is a student that goes out to eat lunch everyday. He eats either at a Korean food place (0), Chinese food place (1), or Burger place (2). The place Kim chooses to go for lunch on day n can be modeled as a DTMC with transition matrix

$$\begin{pmatrix} 0.3 & 0.3 & 0.4 \\ 1 & 0 & 0 \\ 0.5 & 0.5 & 0 \end{pmatrix}.$$

That means if Kim went to the Korean food place yesterday, he will choose to go to the Burger place today with probability 0.4, also there is a 30% chance he will go to the Chinese food place. Are there any absorption states? Is the DTMC ergodic?

- Colab notebook: CP5 §5 (Kim's lunch: model) `[INCLASS] [full]`

Class problem 5

(a) After several days (assume steady state is reached) if you ask Kim what he had for lunch, what is the probability that he would say "burger"?

(b) If on day $n = 1000$ Kim went to the Korean food place, what is the probability that on day $n = 1001$ he will go to the Chinese food place?

(c) If all the students at Kim's university follow the above Markov chain, in the long run, what fraction of the students going to lunch eat at the Chinese food place?

(d) Continuing with question (c) above, if the average cost per lunch is $4,000$ won in the Burger place, $6,000$ won in the Chinese place and $5,000$ won in the Korean place, what is the average revenue per day per customer from all the three eating places together?

(e) How much does Kim spend on lunch (on average) everyday?

▶ Colab notebook: CP5 §5 (Kim's lunch: questions) `[INCLASS] [full]`

Class problem 5

5. Kim's lunch: which quantity is being asked?

(a) $\lim_{n\to\infty} P(X_n = 2) = \pi_2$ (≈ 0.211)

(b) $P(X_{1001} = 1 \mid X_{1000} = 0) = P(X_1 = 1 \mid X_0 = 0) = 0.3$ — one step, no limit needed

(c) ensemble average $= \pi_1$ (≈ 0.263)

(d) $C(0) = 5{,}000,\ C(1) = 6{,}000,\ C(2) = 4{,}000$: $\sum_i C(i)\,\pi_i$ $(\approx 5{,}053$ won per customer per day)

(e) the same $\sum_i C(i)\,\pi_i$ — Kim's time average equals the population's ensemble average (ergodic chain)

PageRank: the random surfer is a Markov chain

- Surfer on n pages: from page i with k links, follow one uniformly $\Rightarrow$ a DTMC with matrix P; the rank of page j is π_j.
- What breaks ergodicity: dead ends (no links out: mass leaks) and spider traps (closed pages: π concentrates there).
- Taxation with rate $1-\beta$: $G = \beta P + \frac{1-\beta}{n}\,\mathbf{1}\mathbf{1}^\top$ is strictly positive $\Rightarrow$ ergodic, unique π, power iteration converges at rate $\leq \beta$.
- Handout (`Google_PageRank_MC.pdf`, next to this deck): $\pi = (0.4, 0.4, 0.2)$; with the spider trap and $\beta = 0.7$, $\pi \approx (0.26, 0.19, 0.55)$.

▷ Proof 7 in the Companion (exam topic): ergodicity of G and $\|p_{k+1} - \pi\|_1 \leq \beta \|p_k - \pi\|_1$.

▶ Colab notebook: PageRank practicum (link farm, taxation, exam drill) `[INCLASS] [full]`